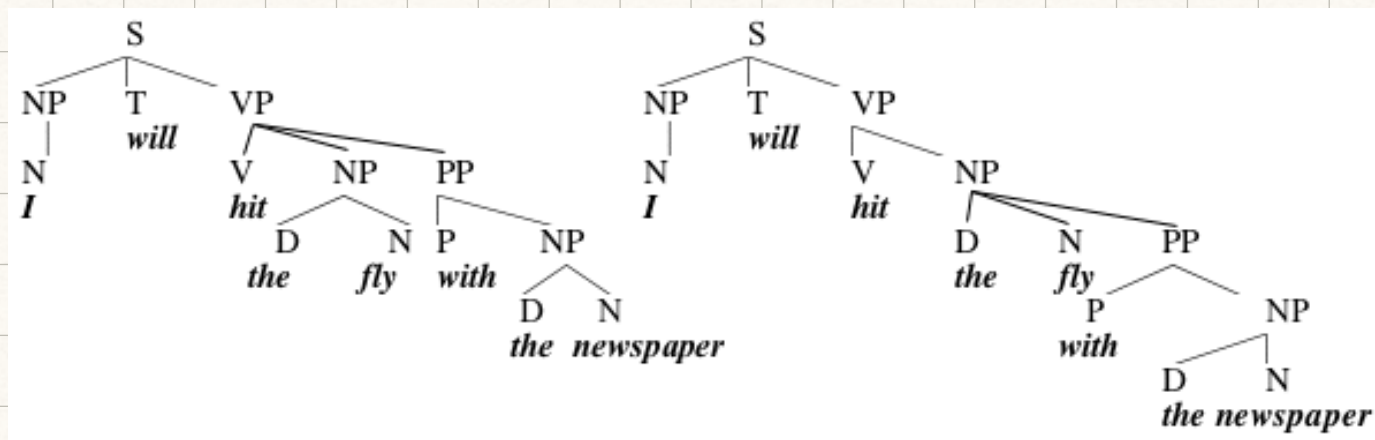
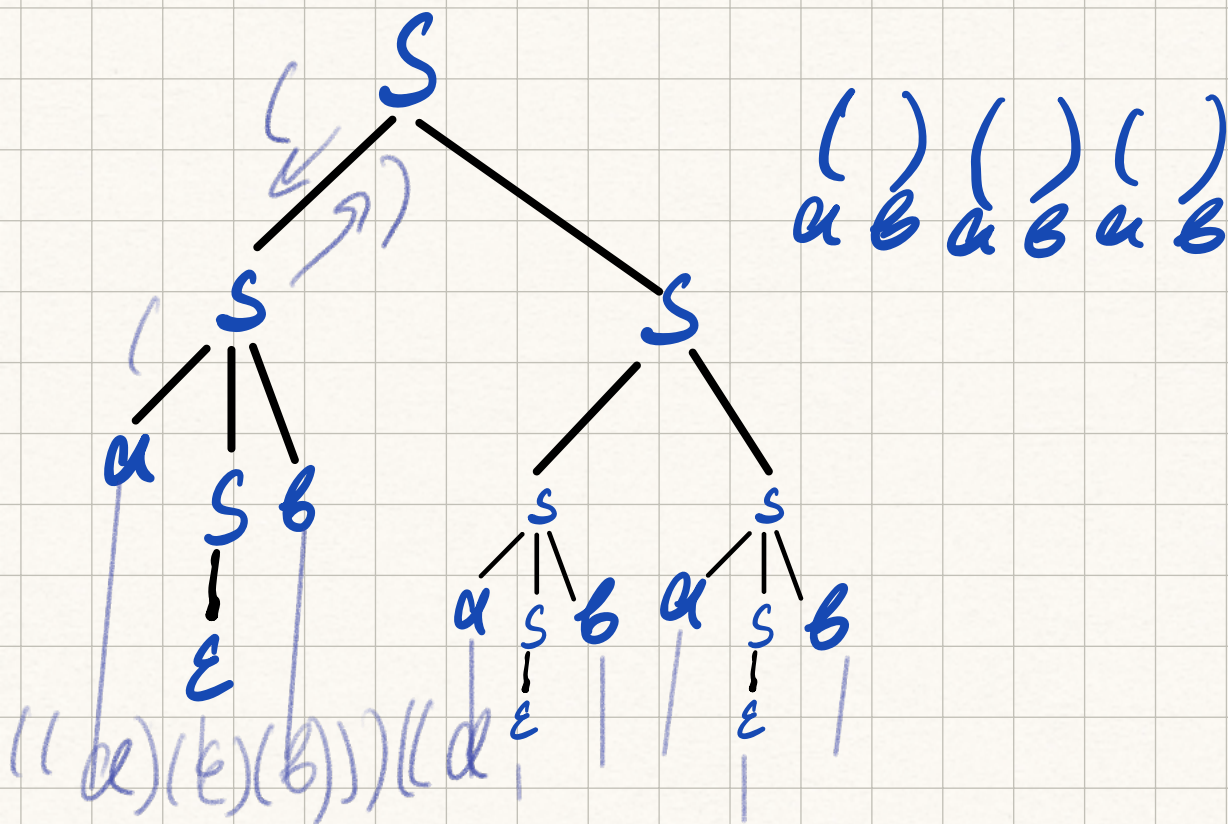


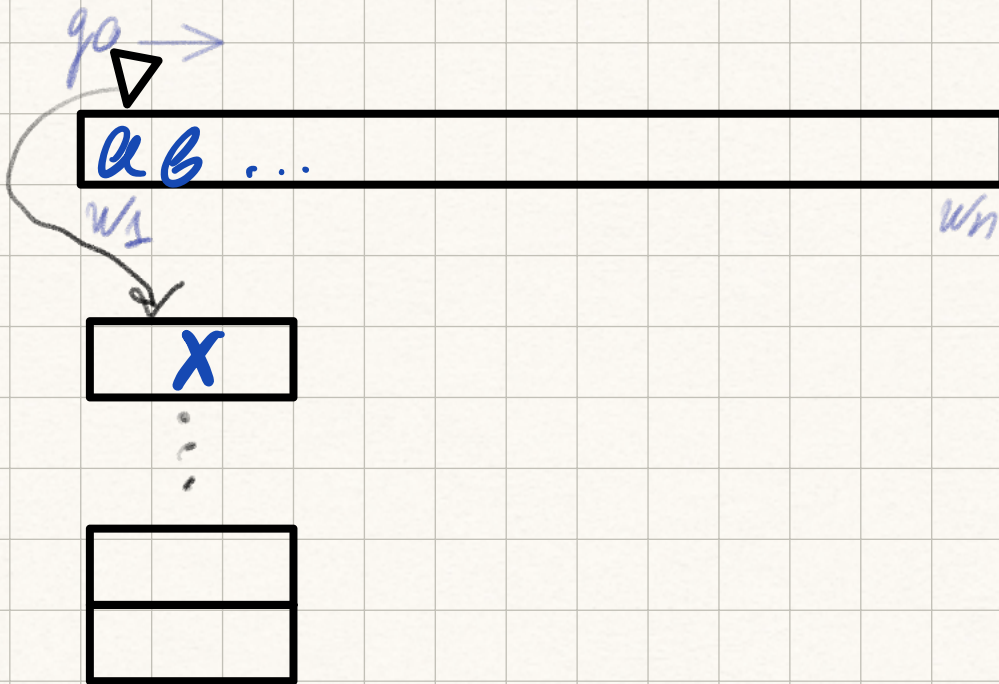
Лекция 2



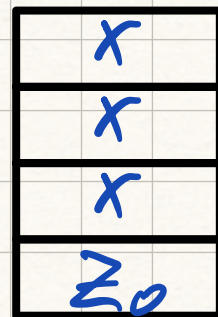
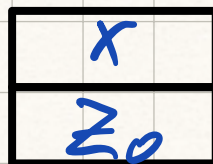
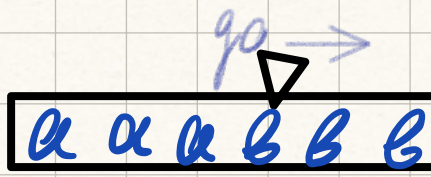
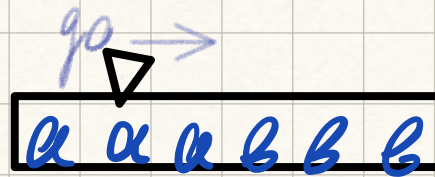
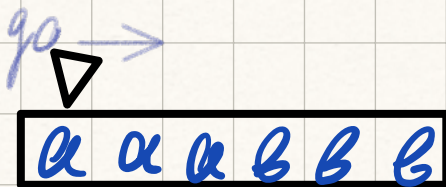
$S \Rightarrow NP VP$



МП - Автоматы

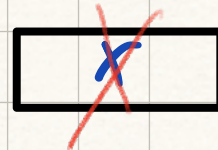
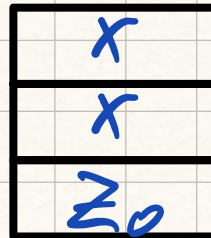
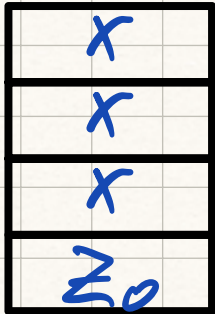
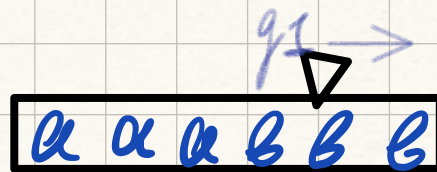
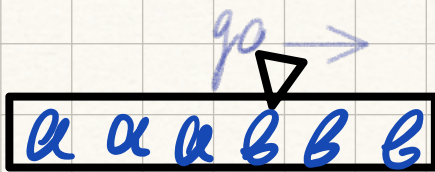


$$L = \{a^n b^n : n \geq 1\}$$

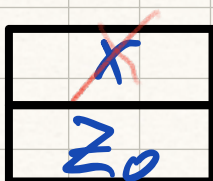
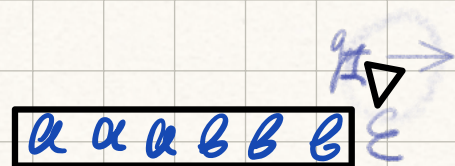
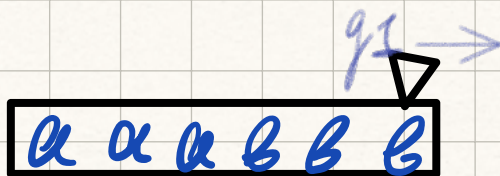


$$(q_0, a, z_0) + (q_0, xz_0)$$

$$(q_0, a, x) \vdash (q_0, xx)$$



$$(q_0, b, x) \vdash (q_1, \epsilon)$$



$$(q_1, \epsilon, z_0) \vdash (q_f, z_0)$$

Определение

$$M(\Sigma, \Gamma, Q, q_0, z_0, F, \delta)$$

Σ → алфавит
 вход
 Γ → алф. стека
 Q → сост.
 q_0 → нач. сост.
 z_0 → прин. сост.
 F → отн. для перехода
 δ

$$Q \times \Gamma^*$$

$$\delta: Q \times (\Sigma \cup \varepsilon) \times \Gamma \rightarrow 2$$

$$(q_0, a, z_0) \vdash (q_0, xz_0)$$

$$(q_0, a, x) \vdash (q_0, xx)$$

$$(q_0, xz_0) \in \delta(q_0, a, z_0)$$

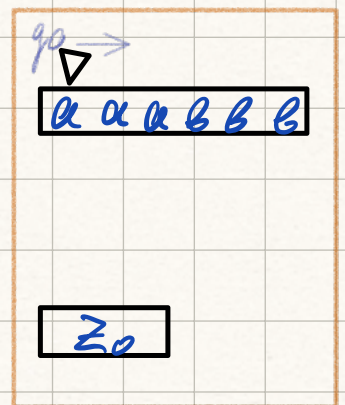
$$\delta(q_0, a, x) = \{ (q_0, xx) \}$$

$$\delta(q_0, a, z_0) = \{ (q_0, xz_0) \}$$

$$(q_0, a, z_0) \vdash (q_0, xz_0)$$

конфигурация:

$$Q \times \Sigma^* \times \Gamma^*$$



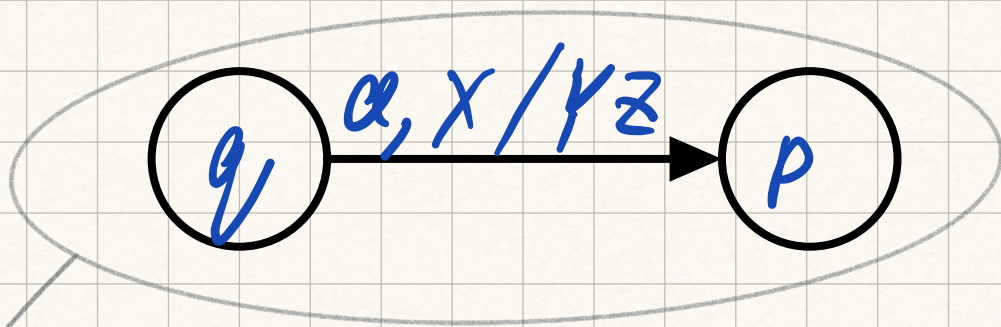
$$(q_0, \underline{a a a b b b}, \underline{z_0}) \vdash$$

$$(q_0, a a b b b, x z_0)$$

↓ перемещение
записи

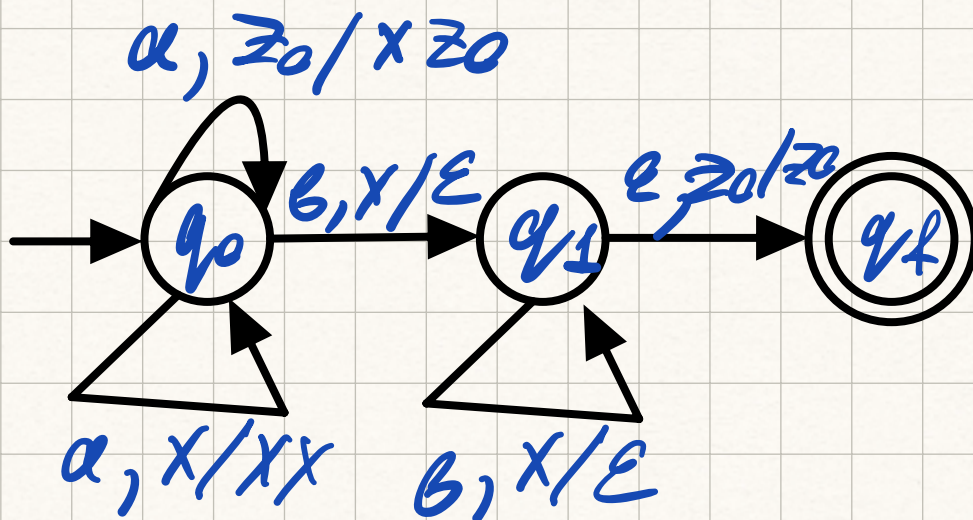
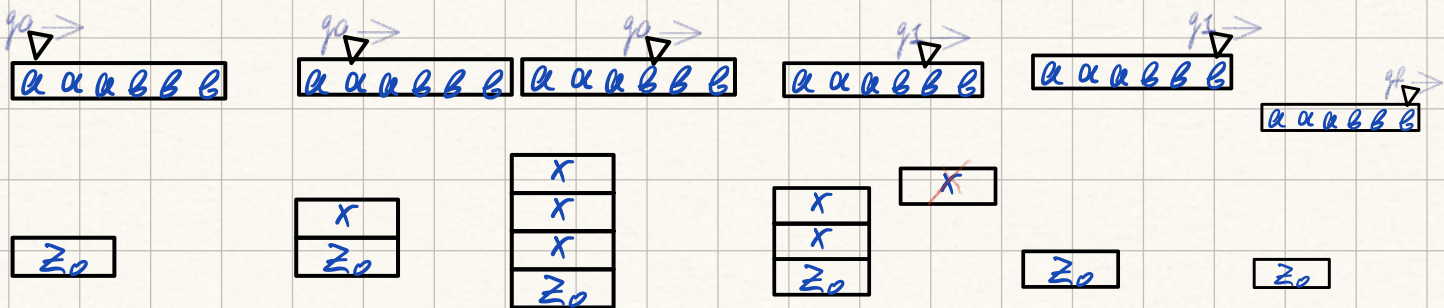
$$(q_0, w, z_0) \vdash^* (q_f, \varepsilon, d)$$

$$q_f \in F, d \in \Gamma^*$$



$\delta(q, a, x) \ni (p, yz)$

$(q, a, x) \quad (p, yz)$



$$\{a^n b^n : n \geq 1\}$$

$$(q_0, a^n b^n, z_0) \vdash^* (q_0, b^n, x^n z_0) \\ \vdash^* (q_1, \epsilon, z_0) \vdash (q_f, \epsilon, z_0)$$

Детерминированные МП-автоматы

$$1. |\delta(q, \underset{\substack{\sigma \in \Sigma \\ \text{символ}}}{\sigma}, x)| \leq 1$$

2. Если $\delta(q, \epsilon, x) \neq \emptyset$,
то $\forall \sigma \in \Sigma$:

$$\delta(q, \sigma, x) = \emptyset$$

МП-автомат детерм.,
если $\forall (q, u, \alpha)$

Доказать из лем., т.е.

$$\exists w: (q_0, w, z_0) \vdash^*$$
$$\vdash^*(q, u, d)$$

$$\text{опр.} \leq \underline{\text{max}}$$

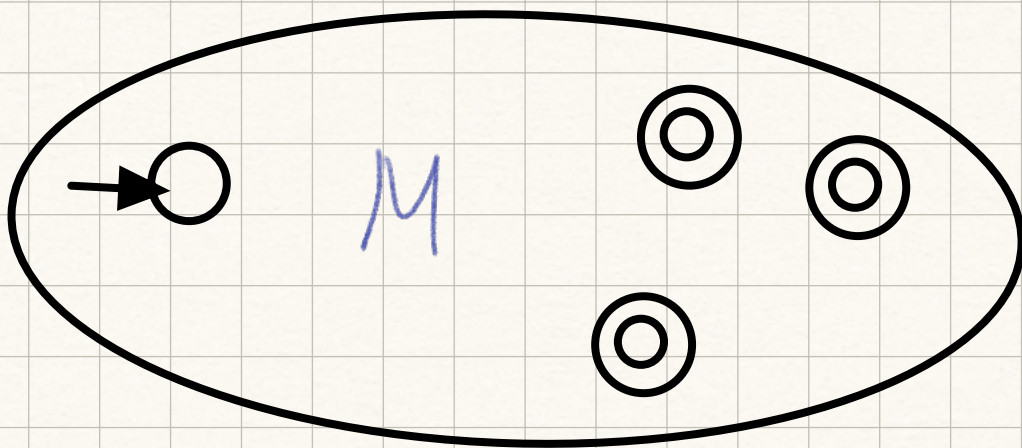
$$(q, u, d) + (q', u', d')$$

$$\text{Если } (q, u, d) + (p, v, \beta),$$
$$\text{то } p = q', v = u', \beta = d'$$

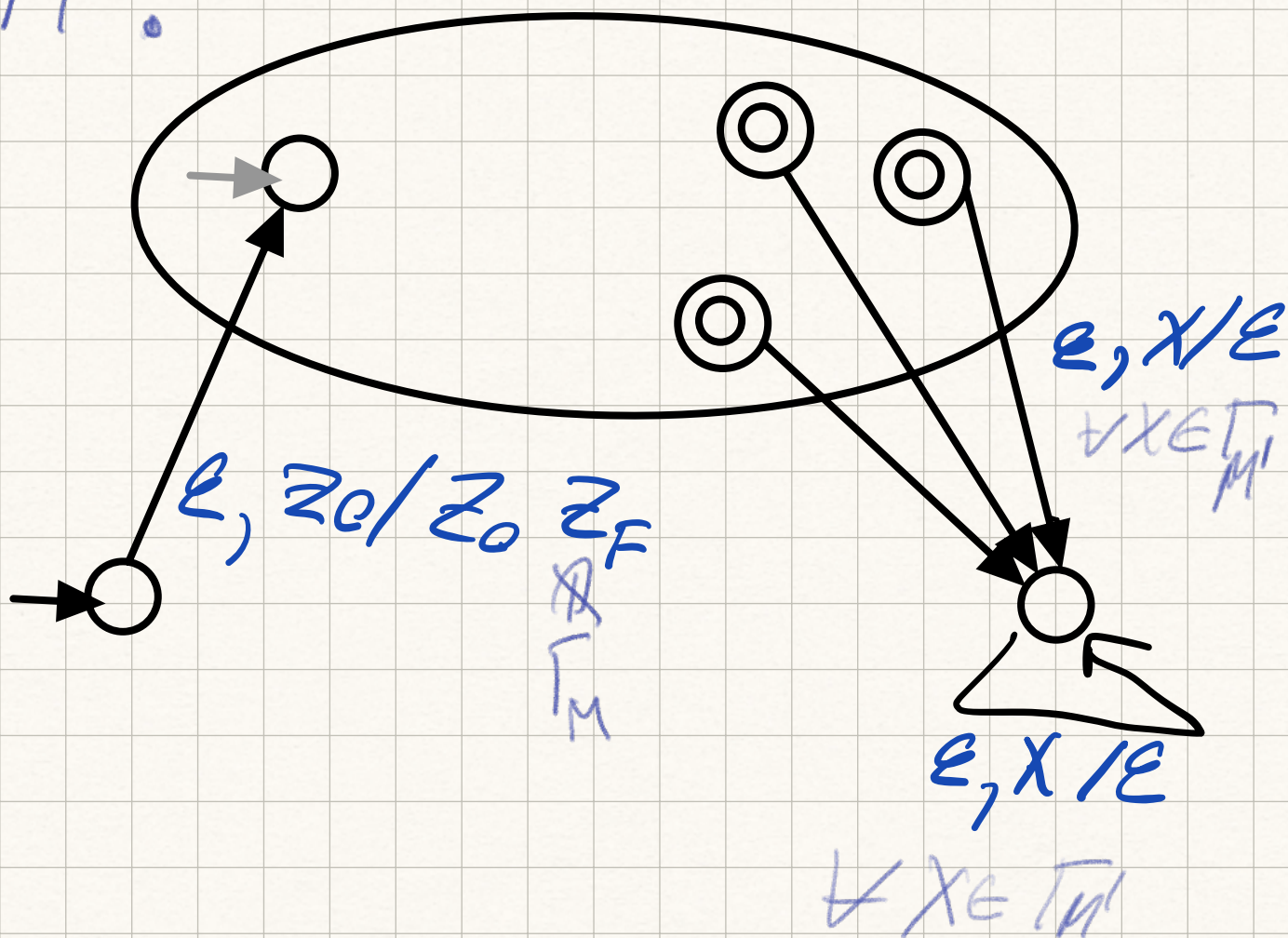
1. Автомат допускает по F
 $(q_0, w, z_0) \vdash^* (q, \varepsilon, z)$

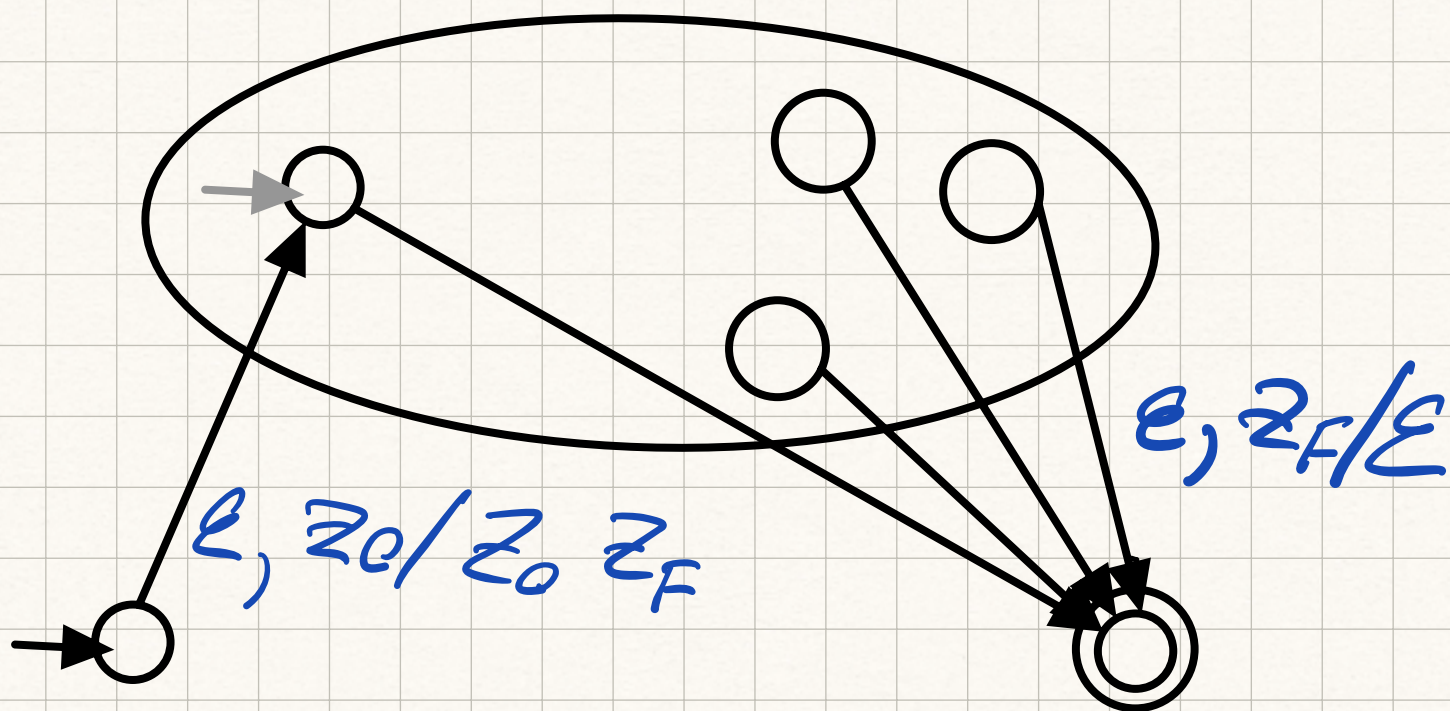
2. Автомат допускает по
нустану стекы

$(q_0, w, z_0) \vdash^* (q, \varepsilon, \varepsilon)$
 $q \in Q$



M' :





M - Д.М.П. - авт. Don.

no нуст. стрек,
 To $w \in L(M)$

$\Rightarrow S \neq \epsilon \Rightarrow wS \notin L(M)$

Упр.

$L(M') =$
 $L(M) \neq$

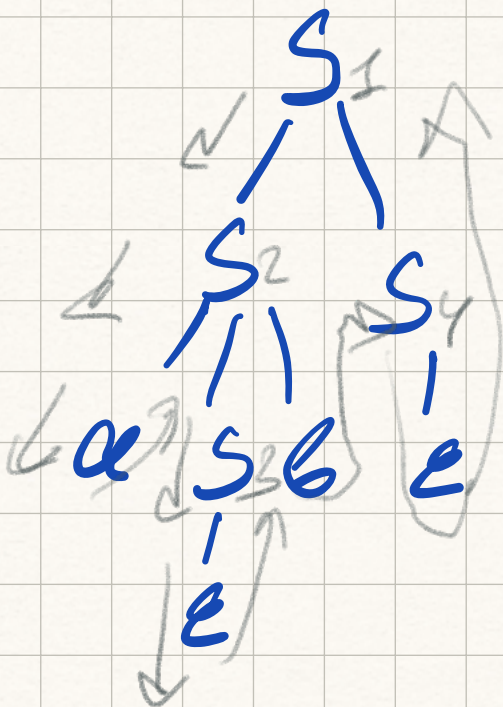
M - Д.М.П. - авт., Don. no
 нпу H соот., то $\exists M'$.
 Д.М.П. - авт. no нуст. Л.

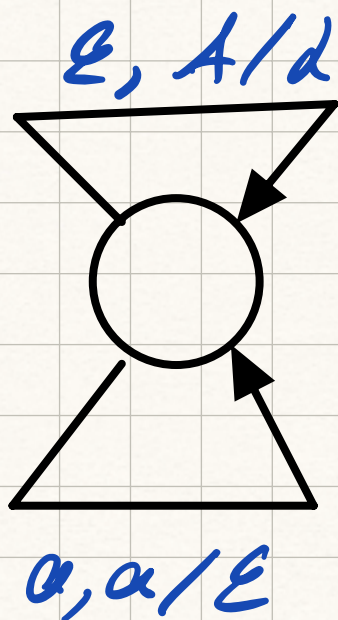
✓ $G \in \text{C.F.G.} \exists M:$

$$L(G) = L(M)$$

$$S \rightarrow aSb \mid SS \mid \epsilon$$

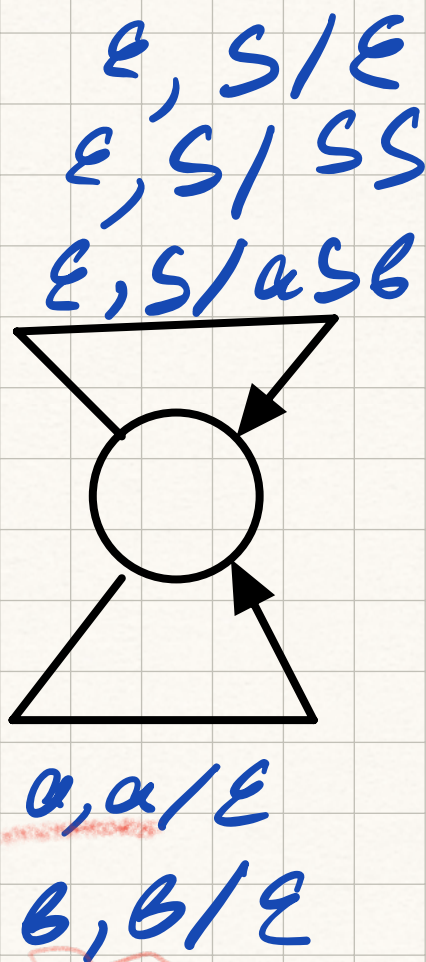
$$S \Rightarrow SS \Rightarrow \underbrace{aSb} S \Rightarrow \underbrace{a} \underbrace{b} \underbrace{\epsilon} S \Rightarrow a b \epsilon$$





$$A \rightarrow d$$

$$Z_0 = S$$



$$S \rightarrow aSb / SS / \epsilon$$

$$S \Rightarrow SS \Rightarrow \underline{aSb}S \Rightarrow \underline{aSb}S \Rightarrow aSbS$$

$$\Rightarrow aSbS$$

$$(q_0, aSb, S) \vdash$$

$$(q_0, aSb, SS) \vdash$$

$$(q_0, \cancel{aSb}, \cancel{aSbS}) \vdash$$

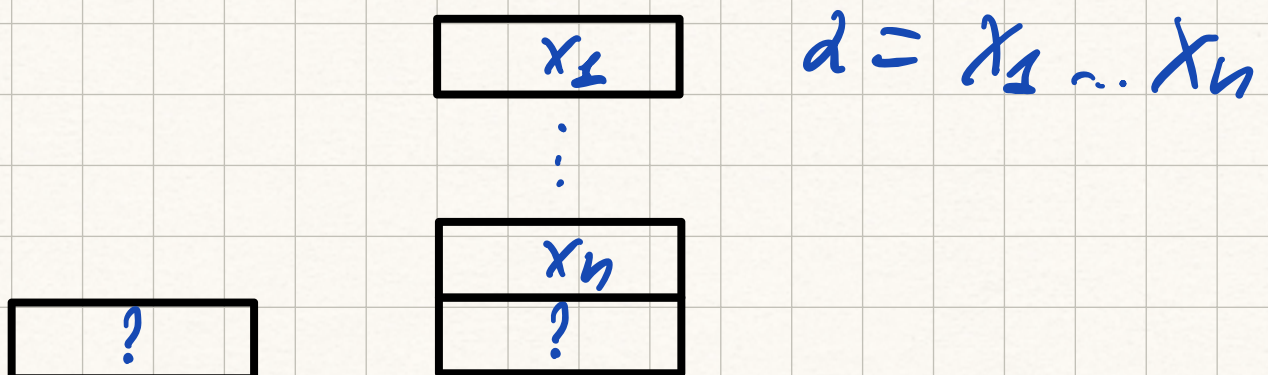
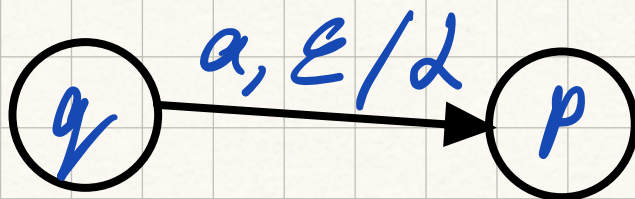
$$(q_0, b, S \cancel{bS}) \vdash (q_0, b, \cancel{bS})$$

$$\vdash (q_0, \epsilon, S) \vdash (q_0, \epsilon, \epsilon)$$

М.П. $M \rightarrow K.C.T \ G.$

1. M движется по
а) пуст. стек
б) по ир. сост.
сдвигу.

2. ϵ ир. по пуст. стек



$$\forall x \in \Gamma: q \xrightarrow{a, x/\alpha x} p$$

$$(q, \alpha u, \varepsilon) \vdash (p, u, \alpha)$$

$$q \xrightarrow{a, \varepsilon/\alpha} p$$

3. За шаг равно 1

символ

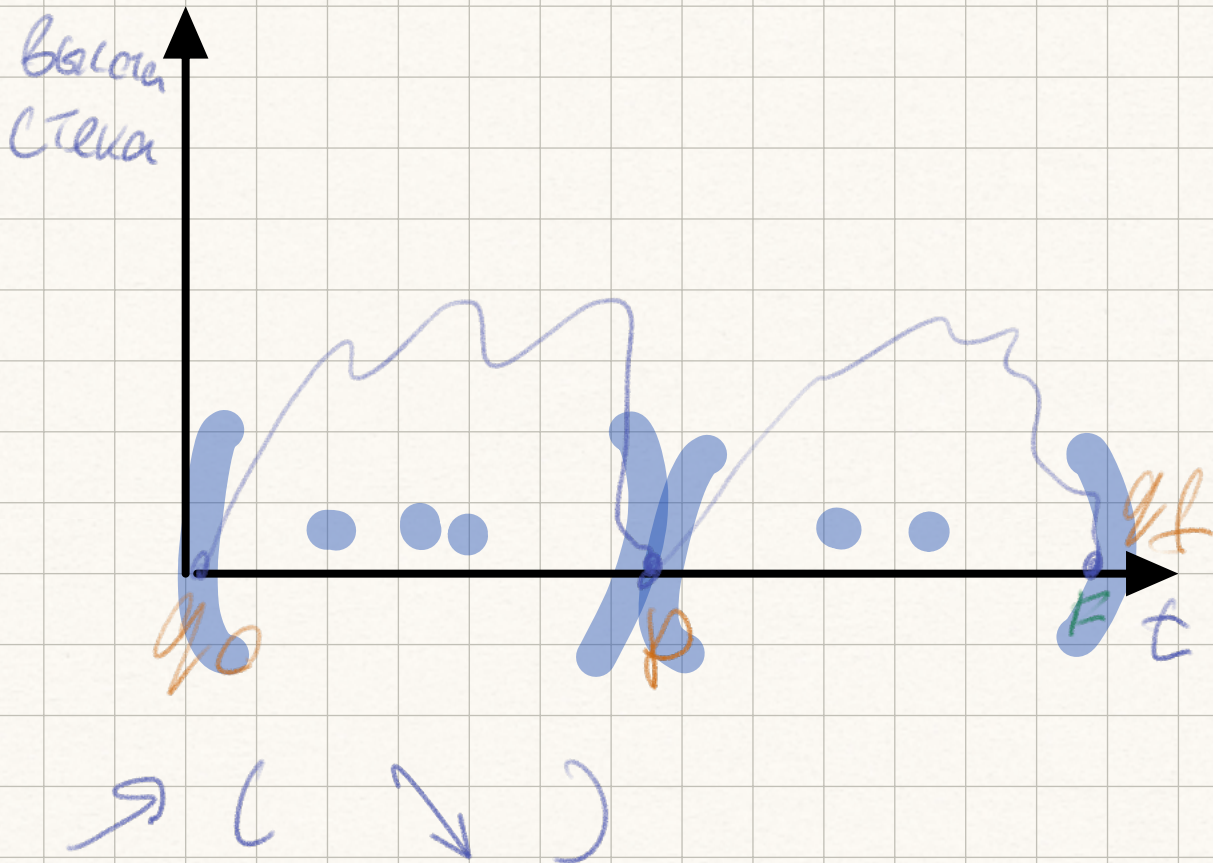
либо добавл. в стек

либо извл. из стека

(либо не удаляется)

$$q \xrightarrow{a, X/K_1 \dots K_n} p$$

$$q^0 \xrightarrow{\alpha, X/\varepsilon} q^1 \xrightarrow{\varepsilon, \varepsilon/\psi_k} \dots \rightarrow q^{k-1} \xrightarrow{\varepsilon, \varepsilon/\psi^I} q^k$$



$$S \Rightarrow A_{\text{top}} A_{\text{ppl}}$$

Балон
Стена

$$\sigma(q_0, a, \varepsilon) = (q', X)$$

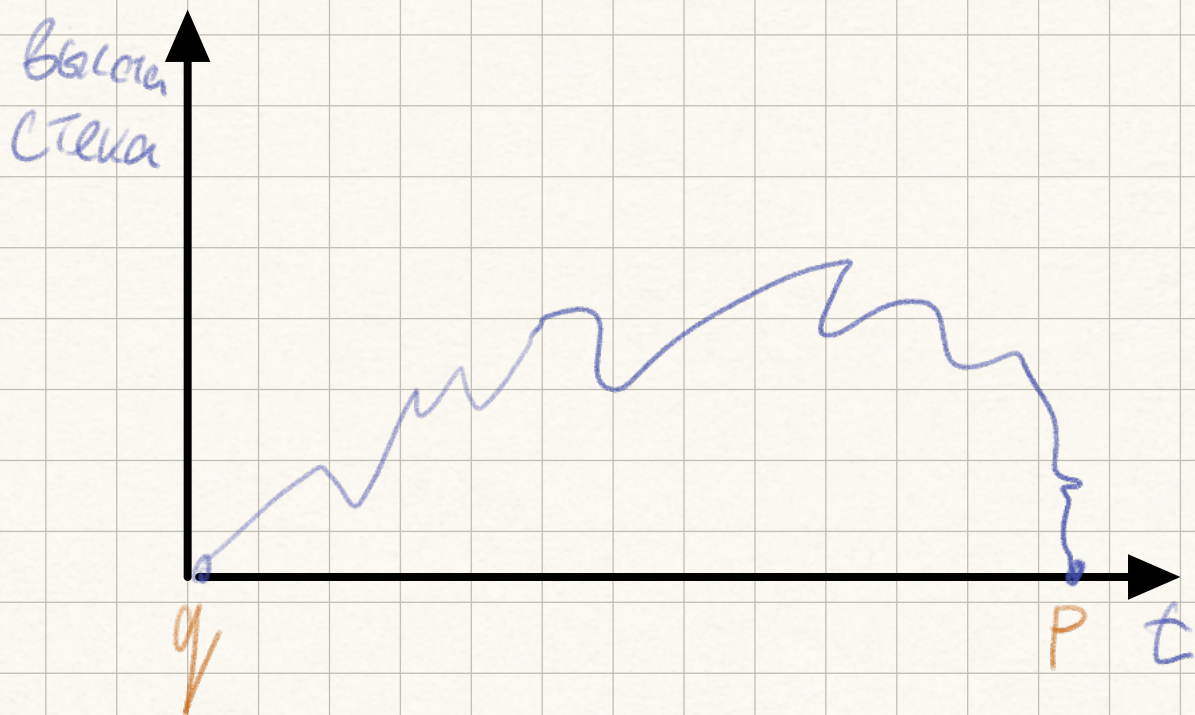
$$\sigma(p', b, X) = (q'', \varepsilon)$$



$$S \Rightarrow \alpha Aq|p|b$$

$$L(A_{qp}) = \{ \mu : (q, \mu, \varepsilon) \xrightarrow{*} (p, \varepsilon, \varepsilon) \}$$

$$\mu : A_{qp} \xrightarrow{*} \mu$$



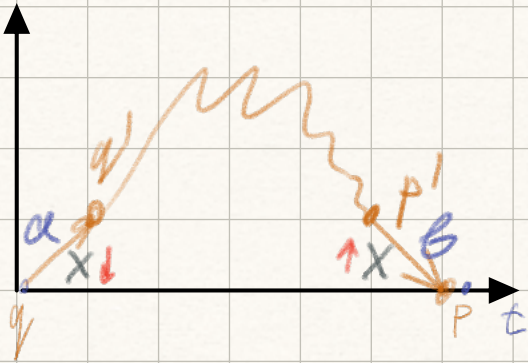
$$S \rightarrow A q_0 q_{F_1} | \dots | A q_0 q_{F_m}$$

$$F = \{q_{F_1}, \dots, q_{F_m}\}$$

$$A_{pp} \rightarrow \varepsilon$$

$$\begin{aligned} a, b &\in \Sigma V \text{ and} \\ q, q', p, p' &\in Q \end{aligned}$$

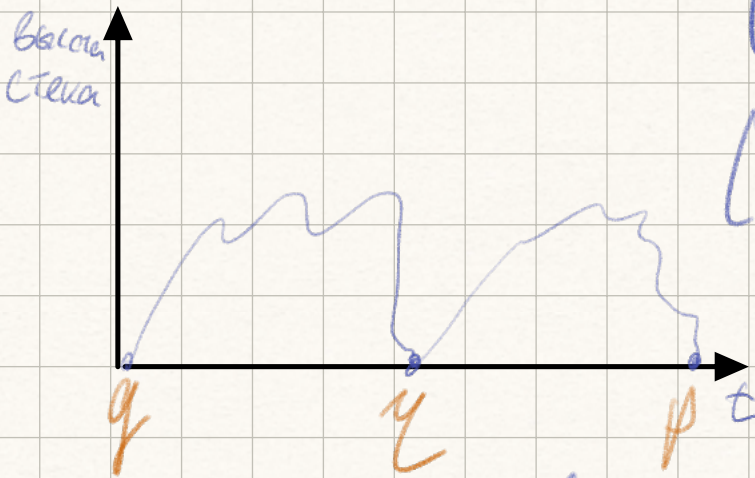
$$\begin{aligned} & \delta(q, a, \varepsilon) \vdash (q', x) \quad x \in \Gamma \\ & \delta(p', b, x) \vdash (p, \varepsilon) \end{aligned}$$



$$A_{g,p} \rightarrow \alpha A_{g',p'} \in$$

$$(q, u, \varepsilon) \neq (r, \varepsilon, \varepsilon)$$

$$(r, v, \varepsilon) \neq (p, \varepsilon, \varepsilon)$$



$$(q, uv, \varepsilon) \vdash^* (p, \varepsilon, \varepsilon)$$

$$A_{qP} \rightarrow A_{qz} A_{zP}$$

$$L(A_{q,p}) = \{u: (q, u, \varepsilon) \vdash^* (p, \varepsilon, \varepsilon)\}$$

$\subseteq$: инд. по длине ввр.
 $n=1$: $A_{q,p} \Rightarrow \varepsilon$

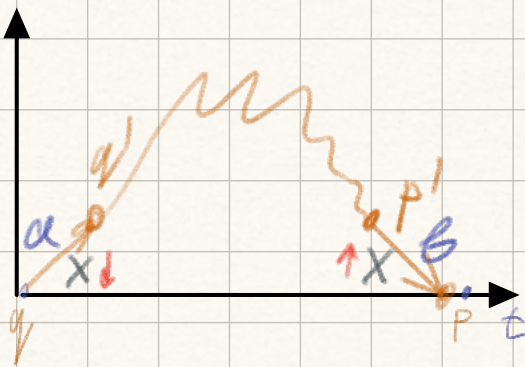
(I)

$$a, b \in \Sigma \cup \varepsilon$$

$$q, q', p, p' \in Q$$

$$\delta(q, a, \varepsilon) \vdash (q', x) \quad x \in \Gamma$$

$$\delta(p', b, x) \vdash (p, \varepsilon)$$



n

$$A_{q,p} \Rightarrow u \quad A_{q',p'} v \Rightarrow u \overset{*}{\Rightarrow} v \quad n-1$$

$$(q, u \cup v, \varepsilon) \vdash (q', u \cup v, x)$$

$$Aq|p| \stackrel{*}{\Rightarrow} u$$

$$\text{и н } \emptyset: (q', u, \varepsilon) \vdash^* (p', \varepsilon, \varepsilon)$$

$$(q, uuv, \varepsilon) \vdash (q', uv, x) \vdash^* (p', v, x) \vdash (p, \varepsilon, \emptyset)$$

$$(q, uuv, \varepsilon) \vdash^* (p, \varepsilon, \varepsilon)$$

$$Aq|p| \stackrel{n}{\Rightarrow} Aq|u| \quad Aq|p| \stackrel{*}{\Rightarrow} u|v|$$

$$k \downarrow^* u \quad \downarrow^* v \quad k < n \quad v < n$$

$$Aq|u| \stackrel{*}{\Rightarrow} u \quad (\Rightarrow) (q, u, \varepsilon) \vdash^* (u, \varepsilon, \varepsilon)$$

$$Aq|v| \stackrel{*}{\Rightarrow} v \quad (\Rightarrow) (u, v, \varepsilon) \vdash^* (p, \varepsilon, \varepsilon)$$

$$(q, uV, \varepsilon) \vdash^* (q, V, \varepsilon) \vdash^* (p, \varepsilon, \varepsilon)$$

$$\supseteq$$

$$n \geq 0$$

$$A_{pp} \Rightarrow \varepsilon$$

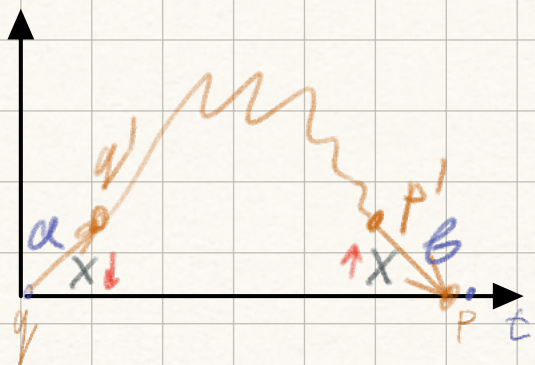
$$L(A_{qp}) = \{u : (q, u, \varepsilon) \vdash^* (p, \varepsilon, \varepsilon)\}$$

$$a, b \in \Sigma \cup \{ \varepsilon \}$$

$$q, q', p, p' \in Q$$

$$\delta(q, a, \varepsilon) \vdash (q', x) \quad x \in \Gamma$$

$$\delta(p', b, x) \vdash (p, \varepsilon)$$



$$(q, auv, \varepsilon) \vdash (q', uv, \varepsilon) \vdash^*$$

$$\vdash^* (p', v, \varepsilon) \vdash (p, \varepsilon, \varepsilon)$$

$$A_{q'p'} \stackrel{*}{\Rightarrow} u$$

$$A_{g,p} \Rightarrow \alpha A_{g',p'} \in \overset{x}{\Rightarrow} \alpha \alpha \in$$

Структурные свойства К.С. языков

$$X, Y \in \text{C.F.L} \quad S \rightarrow S_X \cdot S_Y$$

$$\bullet X \circ Y \in \text{C.F.L}$$

$$\bullet X \cup Y \in \text{C.F.L} \quad S \rightarrow S_X \mid S_Y$$

$$\bullet X \cap Y \not\in \text{C.F.L.} \quad \text{не всегда}$$

$$\overline{X} = \Sigma^* \setminus X \not\in \text{C.F.L.} \quad \text{не всегда}$$

Подстановка

$$\Sigma = \{a_1, \dots, a_n\} \quad L \subseteq \Sigma^*$$

$$L_{a_1}, L_{a_2}, \dots, L_{a_n} \subseteq \Delta^*$$

$$L \xrightarrow{\psi} \bigcup_{L_{a_1}, \dots, L_{a_n} \quad W \in L} L_{w_1} \cdot L_{w_2} \cdot \dots \cdot L_{w_n}$$

$\Delta \quad W = w_1 \cdot w_2 \cdot \dots \cdot w_n$

$$a_i \rightarrow S_{a_i}$$

$$\psi(L)$$

$$X \in C.F.L., R \in REG$$

$$\Rightarrow X \cap R \in C.F.L.$$

$$L(a \cdot b) \rightarrow L_a \cdot L_b \in C.F.L$$

$$(a | b) \rightarrow L_a | L_b \in C.F.L$$

$$a^* \rightarrow L_a^* \in C.F.L.$$

Морфизм: φ

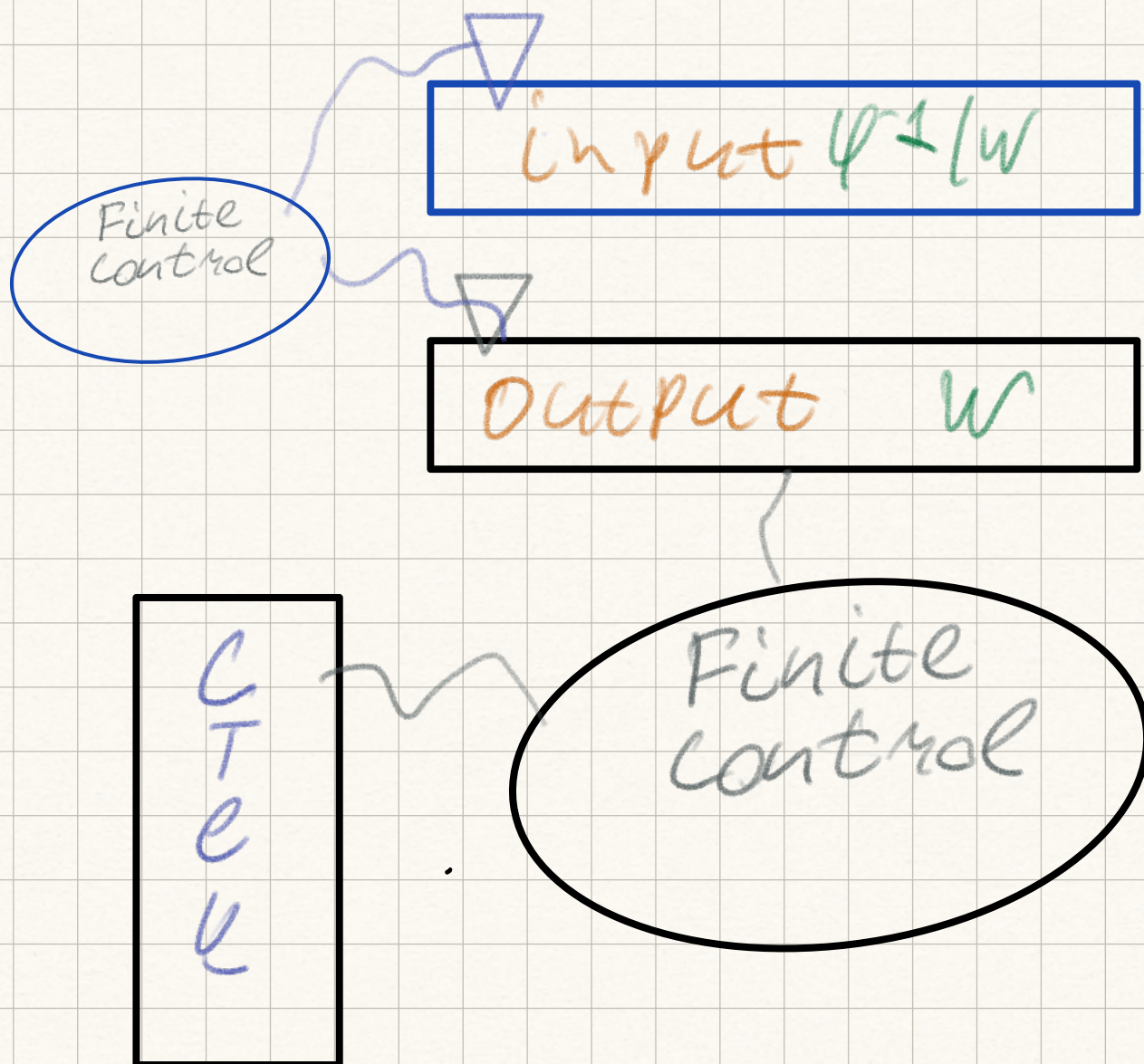
$$\begin{aligned} \varphi(u \cdot v) &= \\ &= \varphi(u) \cdot \varphi(v) \end{aligned}$$

подстановка $a \rightarrow u_a$

$$\varphi(X) \in C.F.L. \quad X \in C.F.L.$$

$$\varphi^{-1}(X) \in C.F.L.$$

$$\varphi(x) = \bigcup_{w \in x} \varphi(w)$$



$\forall X \in \text{C.F.L.} \quad \exists T \in \text{F.S.T.}$
 $T(X) \in \text{C.F.L.}$

Лемма о накачке

$L \in C.F.L. \Rightarrow \exists p \in \mathbb{N}$

$\forall w \in L, |w| \geq p$

$\exists x, u, y, v, z \in \Sigma^*$:

- $w = xuyvz$

$\{a^n b^n c^n : n \geq 0\}$ • $|uyv| \leq p$ $u^0 = \varepsilon$

$\{a^* b^n c^n\}$ • $u, v \neq \varepsilon$

|| • $\forall i \geq 0 \ x u^i y v^i z \in L.$

$L = \{a^n b^n c^n : n \geq 0\}$

$\exists p \Rightarrow a^p b^p c^p = w \in L$

$$|u y v| \leq p \Rightarrow$$

$\Rightarrow u y v$ либо
не содержит a
либо не сод. c

$$\exists x, u, y, v, z \in \Sigma^*$$

- $w = x u y v z$
- $|u y v| \leq p$
- $u v \neq \varepsilon$
- $\forall i \geq 0 \quad x u^i y v^i z \in L.$

$$\underbrace{a^p b^p c^p}_{> p}$$

$$\exists i = 0 : x y z \notin L$$

Зафиксируем $m. \Rightarrow$

$$\left(\begin{array}{l} \# \text{ деревьев, высота} \\ \text{вывода} \end{array} \leq m \right) \leq e^m$$

конечно

→
лемма
самой
длинной
правой части

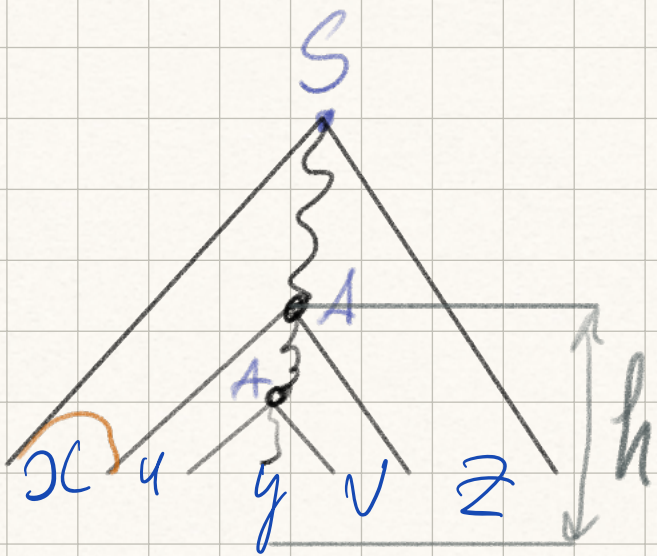
$$\exists p \quad |W| \geq p \Rightarrow$$

$$S \stackrel{t}{\Rightarrow} W \quad t > |N|$$

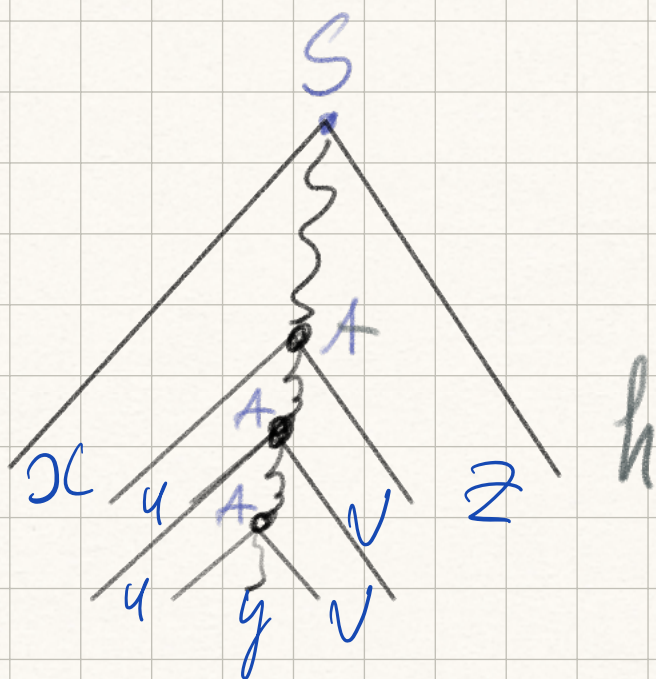
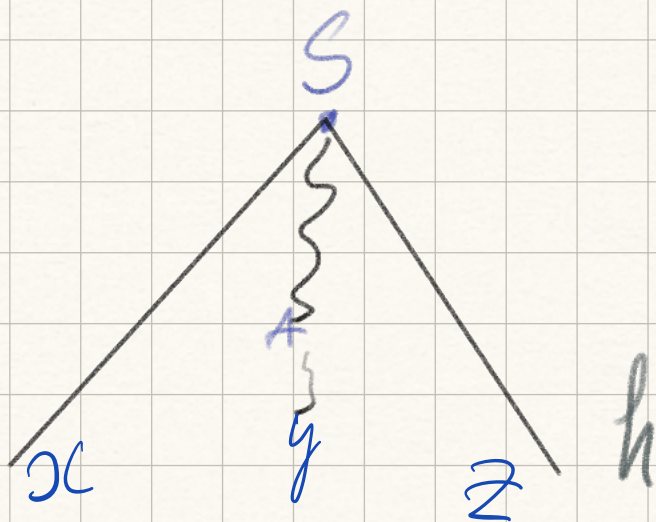
$$p = e^{|N|}$$

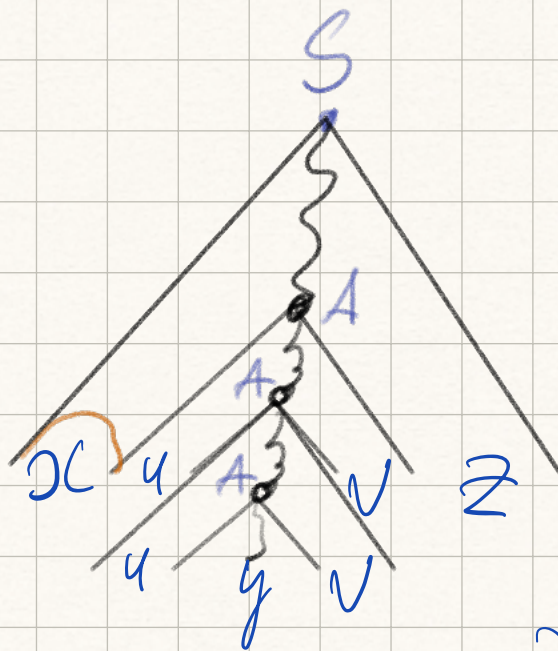
дерево вывода
в форме мнн.

$$\Rightarrow u \cdot v \neq \varepsilon$$

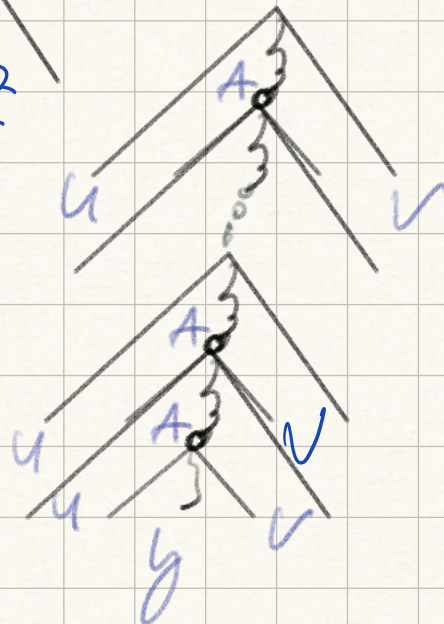
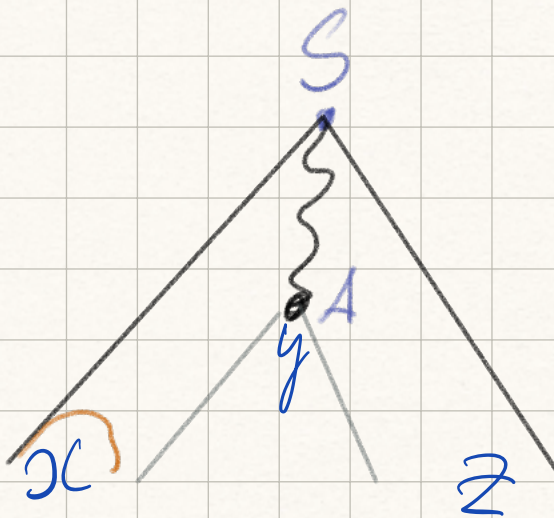


$$h \leq |N| \Rightarrow |u y v| \leq \underbrace{e^{|N|}}_{e^{|h|}} \underbrace{p}_{p}$$





$$x u^i y v^i z$$



$$A \Rightarrow x_1 x_2 \dots x_n \quad x_i \in NUTV \cup \{e\}$$

1. Если $S \rightarrow \epsilon \Rightarrow S$ не вх. в
нр. часть нр.

2. $A \rightarrow BC$
 $A \rightarrow a$) каждое правило
имеет такой
вид.

Нормальная форма Хомского

1. Укор. правила

$A \rightarrow X_1 X_2 \dots X_n$

$X_i \in \Sigma \cup N$

$A \rightarrow [X_1 \dots X_{n-1}] [X_n]$

→ новый нестерм.

$X_n \in N$

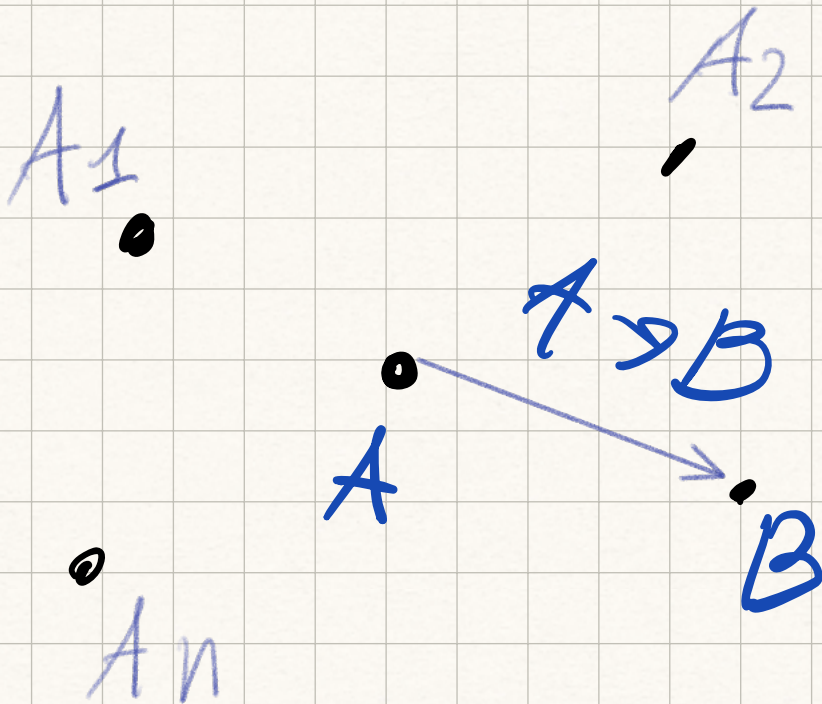
$[a] \rightarrow a$

$[a]$

$$[x_1 \dots x_k] \rightarrow [x_1 \dots x_{k-1}][x_k]$$

2. Цепные правила

$$A \rightarrow B$$



C дост. из A

$$C \rightarrow x_k$$

$$A \rightarrow x_k$$

$$A: A \stackrel{*}{\Rightarrow} \epsilon$$

$$A \rightarrow B'C'$$

$$B \stackrel{*}{\Rightarrow} \epsilon$$

$$\hookrightarrow A \Rightarrow C$$

$$C \stackrel{*}{\Rightarrow} \epsilon$$

$$A \Rightarrow B$$

* Нуксен ал-и
бадалетия кебест.
самб.